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1.Introduction

Real Time Collaboration Tool is a web-based platform and service that provides collaboration features for software development projects. In this there is a feature of creating projects, workspaces that provides services for selecting the user for assigning the workspace and generating the report, managing collaboration and managing task provides the feature of directly assigning the task to the user. It serve as a centralized hub where team members can collaborate, exchange information, and work together efficiently, ultimately leading to improved productivity and collaboration within an organization. It allows developers to manage the task easily and also making it easier to collaborate with team members and maintain a well-organized history of project development.

1.1 Existing System

It is an open-source platform that provided collaboration features for software development projects in which anyone can access the repository and do the changes in the repository. In this project manager can create repositories after that for assigning them he has to select the developer from the list, admin create the project after that he has view the list of project managers from that it will select the project manager for the project and for assigning the task to the user after adding in the list project manager has to go to view the list of the user from that he has to select the user for assigning the task. This process is lengthy and time consuming.

1.2 Need for the New System

Real-time collaboration tools have become increasingly essential in today's dynamic and interconnected work environments. With the rise of remote work and distributed teams, there is a growing need for collaboration tools that allow team members to work together seamlessly, regardless of their physical locations. In a workspace or project environment, team members or project managers can create and delegate tasks or assignments to specific individuals. Project Manager can select the intended recipients from a list of team members or collaborators who are part of the shared workspace or project. This feature streamlines task delegation, making it easy to assign responsibilities, within a collaborative environment. Project Manager can often search for team members or select from a pre-populated list, ensuring efficient and accurate task assignments. It provides smooth integration with other commonly used tools and platforms.

1.3 Objective of the New System

The objective of implementing a new system in a real-time collaboration tool is to enhance and streamline collaborative processes within an organization. The specific objectives may vary depending on the organization's goals and requirements, but some common objectives include:

- i. Improve Communication:
- ii. Enhance Collaboration:
- iii. Increase Productivity
- iv. Facilitate Project Management

1.4 Problem Definition

- **Low and Inefficient Collaboration:** Current collaboration methods are slow and inefficient, causing delays and wasted effort.
- **Delayed Communication:** Lack of instant communication hampers quick decision-making and real-time collaboration.
- **Limited Access and Compatibility:** Tools may not work well across devices or systems, limiting accessibility and flexibility, especially for remote work.
- **Disorganized Workflow:** Existing tools often lack efficient workflow management, leading to confusion and disorganization.
- **Privacy and Security Issues:** Users worry about data privacy and security in the tools they use, affecting trust and adoption.

1.5 Core Components

- **User Authentication and Authorization:** Secure user login and role-based access control to ensure only authorized individuals can access and collaborate within the tool.
- **User Roles and Permissions:** The ability to assign different roles with specific permissions is crucial for controlling access.
- **Task and Project Management:** Tools for creating, assigning, tracking, and managing tasks and projects collaboratively.
- **User Profiles and Collaboration History:** User profiles detailing roles, responsibilities, and collaboration history to enhance understanding of team dynamics and individual contributions.
- **Security and Data Privacy:**
 - User Authentication:** Protecting user accounts from unauthorized access.

1.6 Project Profile

Project Title	Real Time Collaboration Tool
Front-end	HTML5, CSS3, Bootstrap4, Django
Back End	My SQL
Server	Xampp v3.3.0
Other Tools	Power Point, MS Word - 2019
Diagram Builder	draw.io

1.7 Assumptions and Constraints

❖ Assumption:

- Users can register and create an account with a unique username and email.
- User authentication is in place to ensure secure access to the collaboration tool.
- Users can have different roles, such as developers, project managers, and admins, each with their respective permissions and capabilities.
- Collaborations are created by Collaboration Owners, who can invite users to join the collaboration.

❖ Constraints:

- The collaboration tool must be accessible via a web interface or dedicated application.
- The system must handle concurrent user interactions and ensure data integrity in real-time collaboration scenarios.
- User information and collaboration data must be securely stored and protected.
- It may not fully replace the benefits of face-to-face interactions, such as building personal relationships and reading non-verbal cues.

1.8 Advantages and Limitations of the Proposed System

❖ Advantages:

- Enhanced communication and instant information exchange.
- Increased productivity and faster decision-making.
- Enablement of remote work and collaboration across locations.
Simultaneous editing for efficient collaboration

❖ Limitation:

- Dependency on stable internet connectivity and technical infrastructure.
- Security and privacy concerns regarding data protection.
- Disruption to work-life balance with constant access to the tool.

2. Requirement Determination & Analysis

2.1 Requirement Determination:

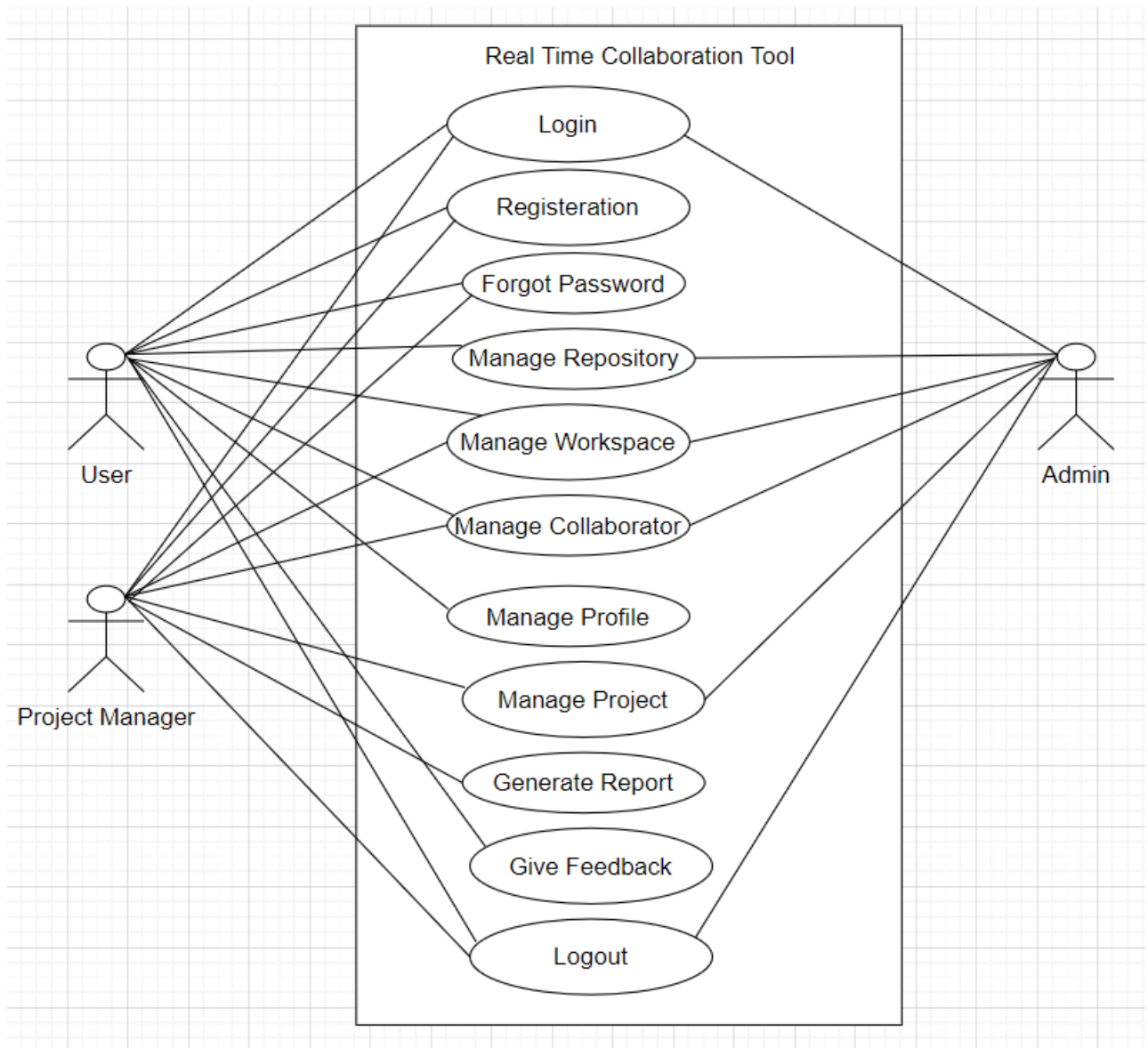
- Understand Users:
 - Identify who will collaborate and how they'll use the system.
- Define Features:
 - Specify needed tools like chat, editing, and file sharing.
- Ensure Technical Compatibility:
 - Check devices, networks, and platform compatibility.
- Focus on Real-Time Updates:
 - Ensure seamless, instant synchronization.
- Prioritize Security:
 - Implement strong data encryption and access controls.
- Consider Scalability:
 - Plan for growth and increased usage.
- Comply with Regulations:
 - Adhere to industry standards and governance policies.
- Optimize UX:
 - Design an intuitive interface for easy collaboration.
- Test Thoroughly:
 - Validate features under various conditions.
- Provide Support:
 - Offer documentation and training for users.

2.2 Targeted Users

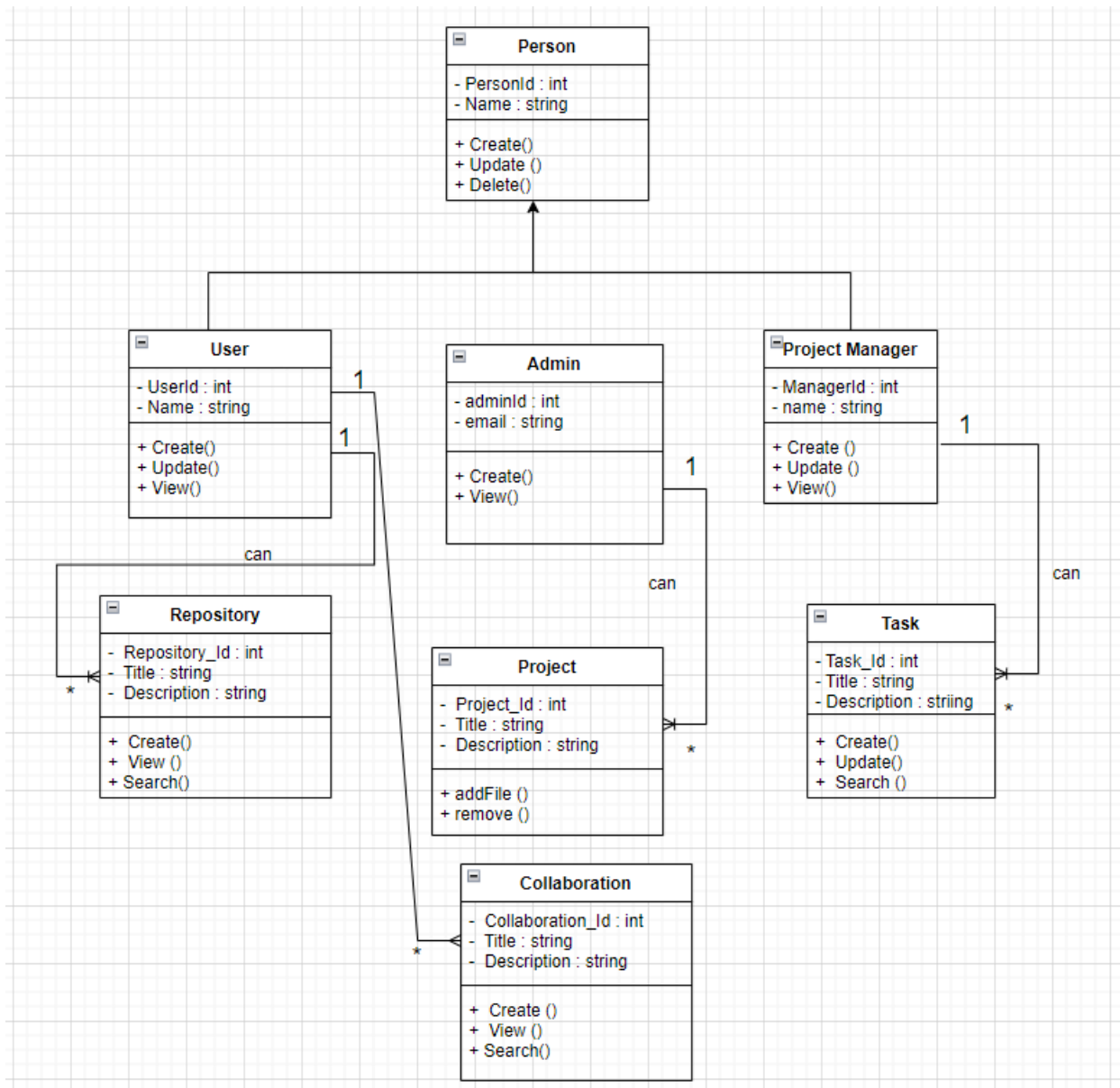
Targeted Users	Details of users
Admin	• Manage Project • View Feedback • Generate Report
Project Manager	• Manage Task • Manage Repository • Mange Workspace • View Collaborators
User	• View Task • View Workspace • Manage Collaborators • Manage Profile

3. System Design

3.1 Use Case Diagram:

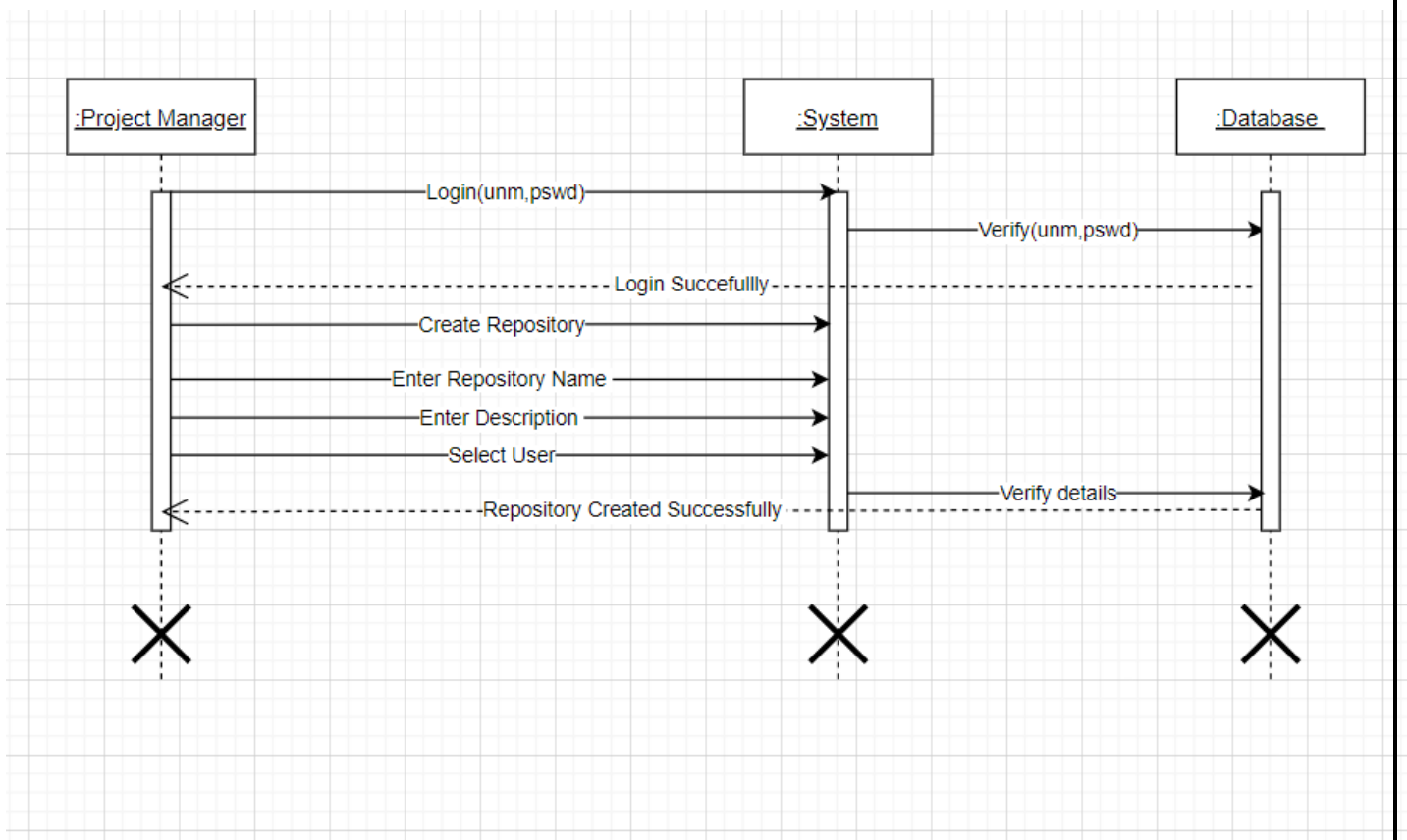


3.3 Class Diagram

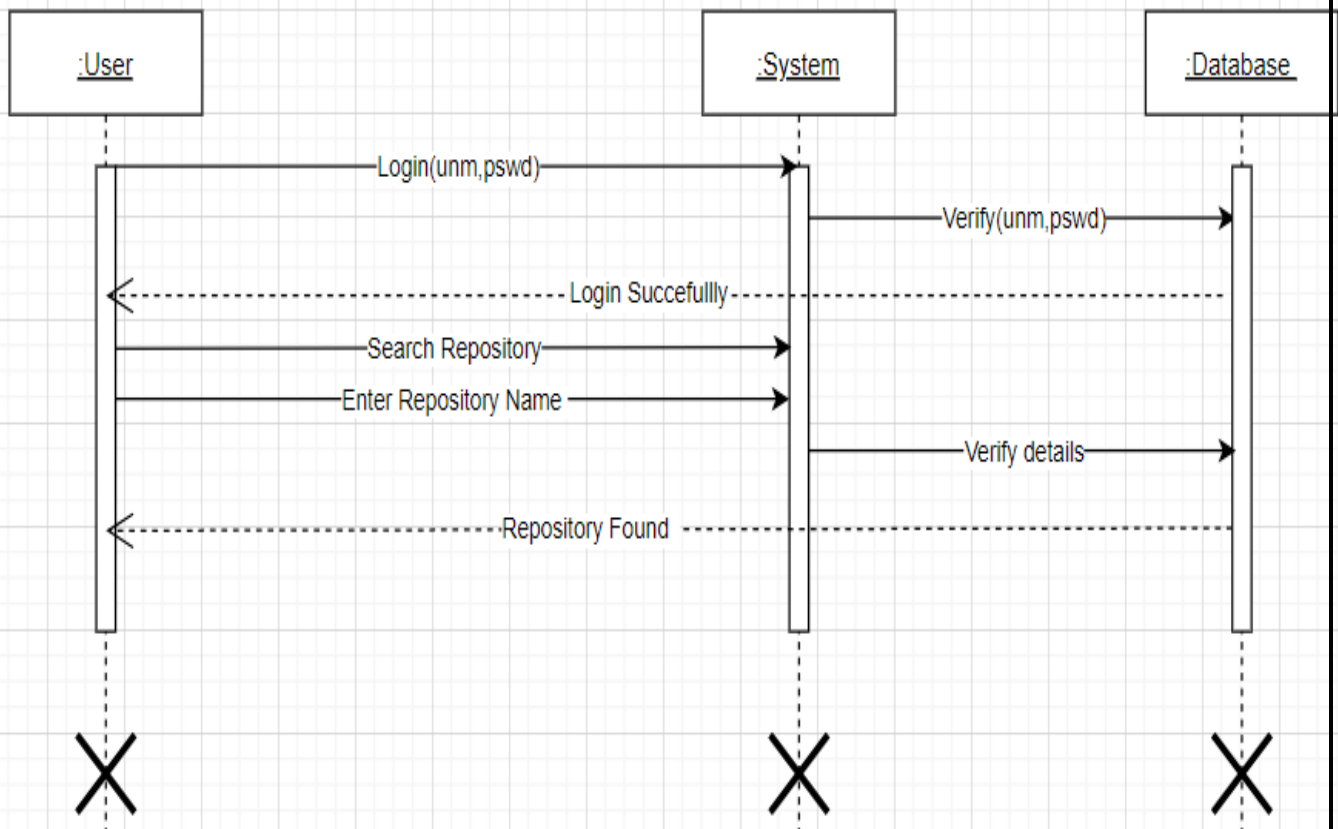


3.3 Interactive Diagram

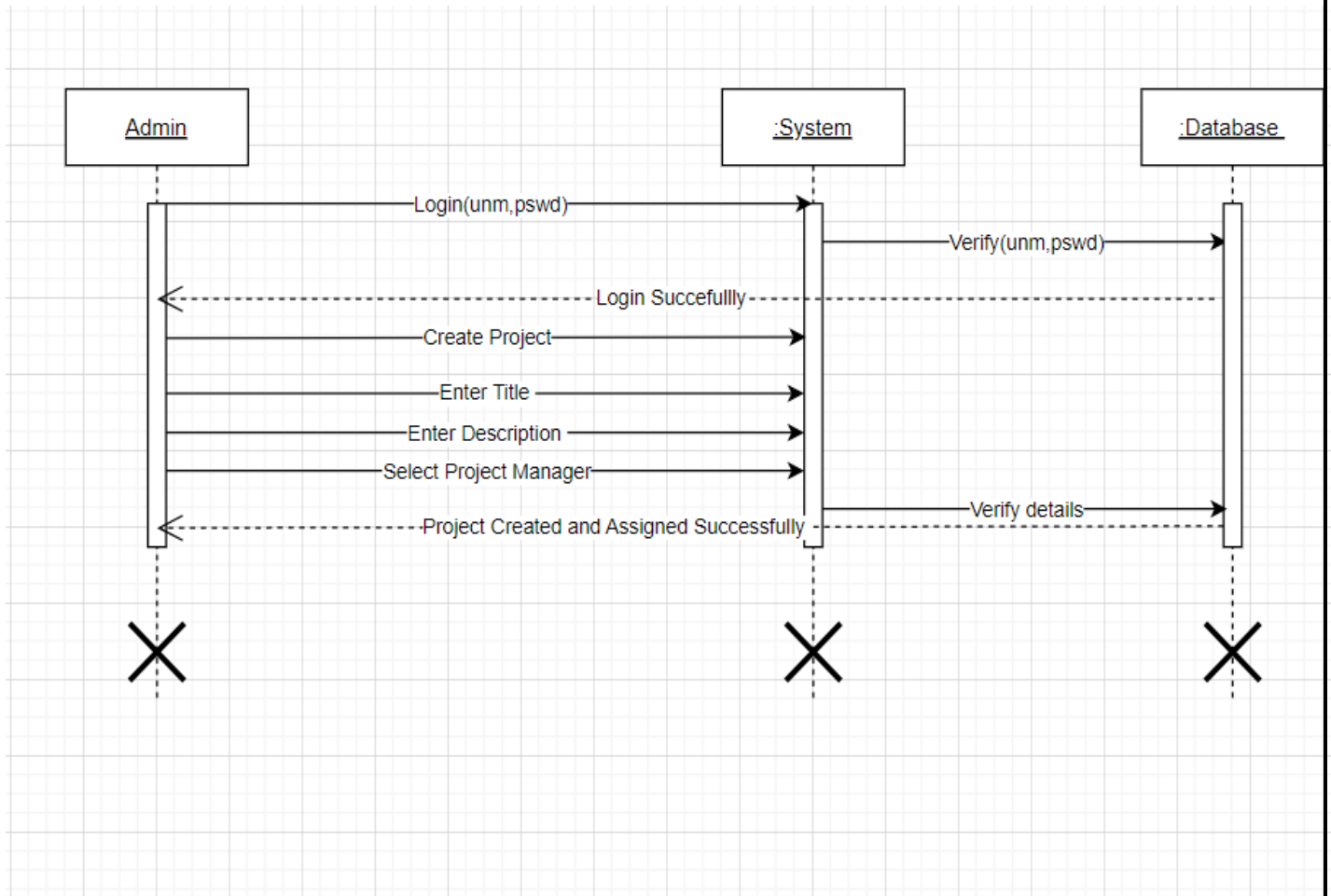
Sequence Diagram to create repository by Project Manager



Sequence Diagram to view repository by User

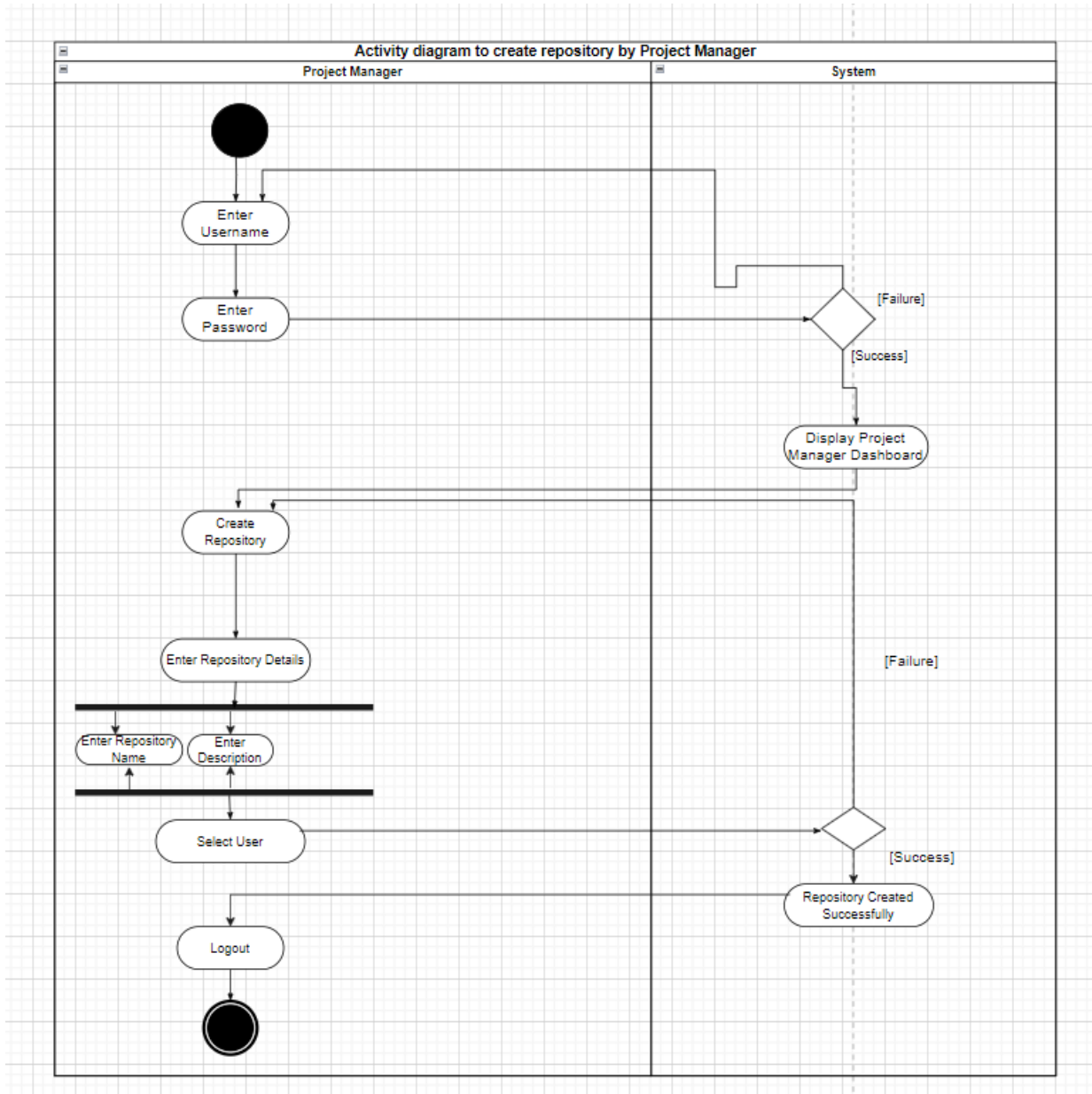


Sequence Diagram to create project by Admin

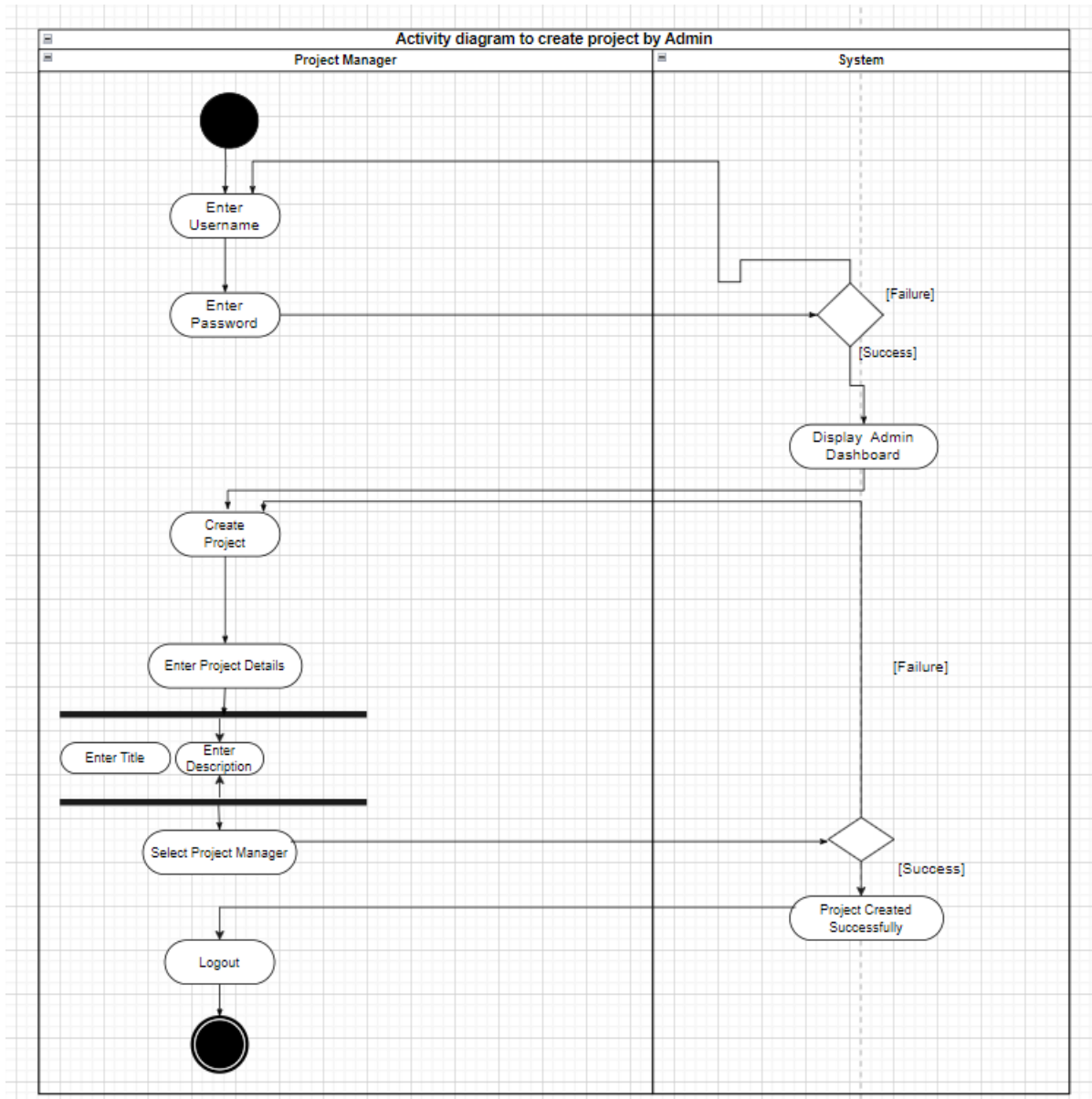


3.4 Activity Diagram

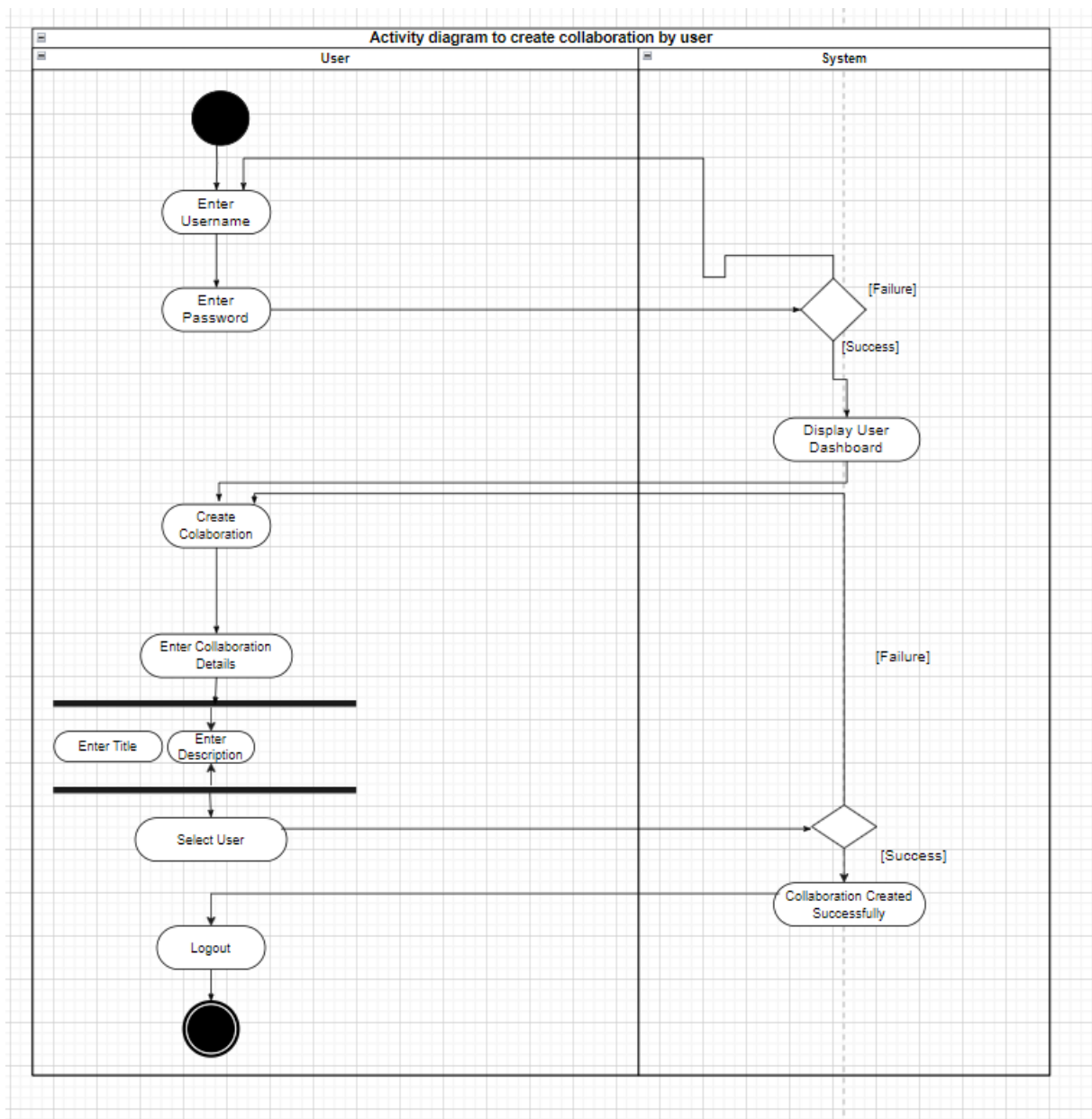
Activity Diagram to create repository by Project Manager:



Activity Diagram to create project by Admin



Activity Diagram to create collaboration by User:



3.5 Data Dictionary

Table: user profile

Table Comments						
Columns						
Name	Data Type	Nullable	PK	FK	Default	Comment
id	BIGINT (20)	No	Yes	No	None	
name	VARCHAR (100)	No	No	No	None	
email	VARCHAR (254)	Yes	Yes	No	None	
user type	VARCHAR (20)	No	No	No	None	
password	VARCHAR (100)	Yes	No	No	NULL	
username	VARCHAR (100)	Yes	Yes	No	NULL	
Language preference	VARCHAR (30)	No	No	No +	None	

Table: collaborator

Table Comments						
Columns						
Name	Data Type	Nullable	PK	FK	Default	Comment
id	BIGINT (20)	No	Yes	No	None	
first_name	VARCHAR (50)	No	No	No	None	
last_name	VARCHAR (50)	No	No	No	None	
email	VARCHAR (254)	No	No	No	None	
role	VARCHAR (50)	No	No	No	NULL	

Table: password token

Table Comments						
Columns						
Name	Data Type	Nullabl e	PK	FK	Defau lt	Com ment
id	BIGINT (20)	No	Yes	No	None	
email	VARCHAR (30)	No	No	No	None	
otp	VARCHAR (8)	No	No	No	None	
timestamp	DATETIME (6)	No	No	No	None	

Table: repository

Table Comments						
Columns						
Name	Data Type	Nullabl e	PK	FK	Defau lt	Com ment
id	BIGINT (20)	No	Yes	No	None	
name	VARCHAR (100)	No	No	No	None	
description	longtext	No	No	No	None	
created_at	date	No	No	No	None	
last_updated_at	date	No	No	No	None	
owner_id	int(11)	No	Yes	No	None	

Table: task

Table Comments						
Columns						
Name	Data Type	Nullabl e	PK	FK	Defau lt	Com ment
id	BIGINT(20)	No	Yes	No	None	
title	VARCHAR(200)	No	No	No	None	
description	LONGTEXT	No	No	No	None	
created_at	DATE	No	No	No	None	
updated_at	DATE	No	No	No	None	
assign_to	VARCHAR (100)	No	Yes	No	None	

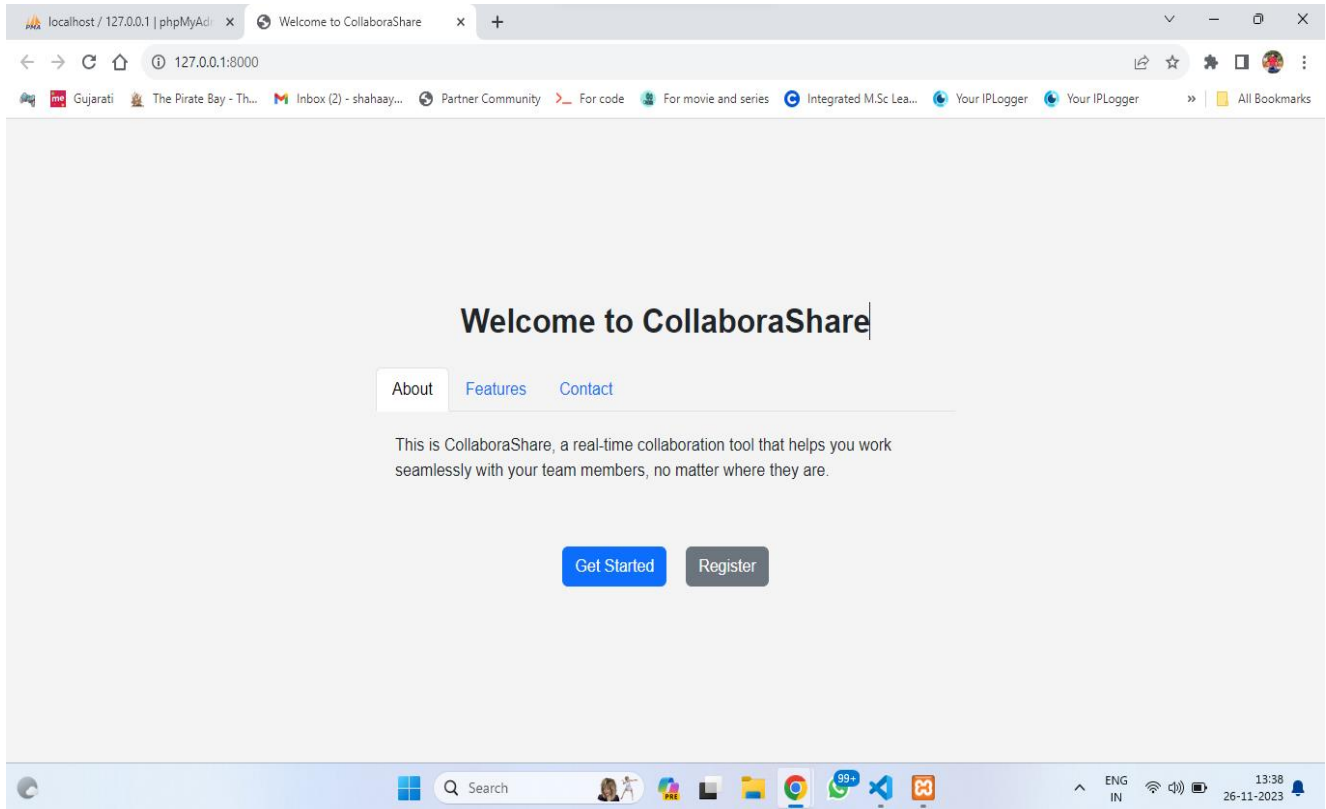
4. Development

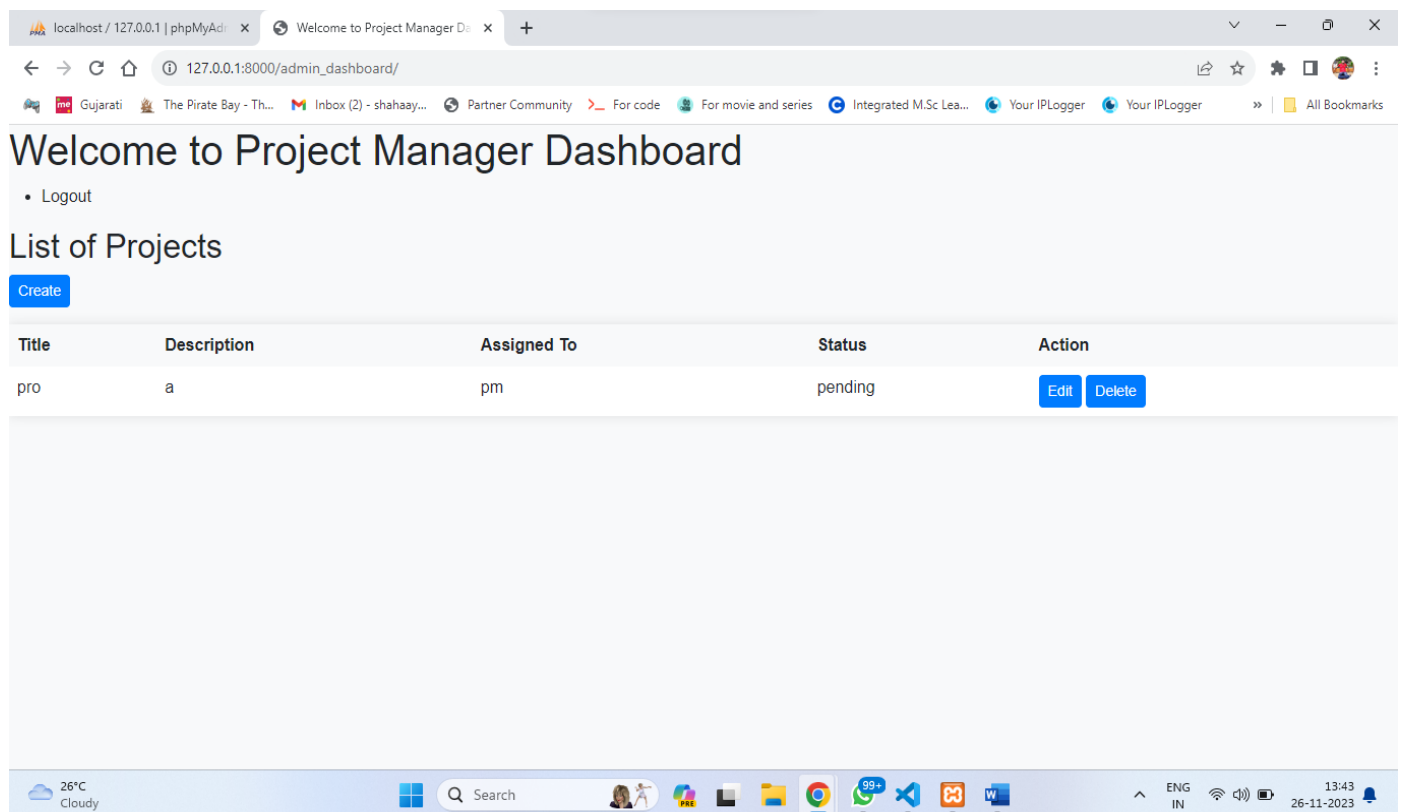
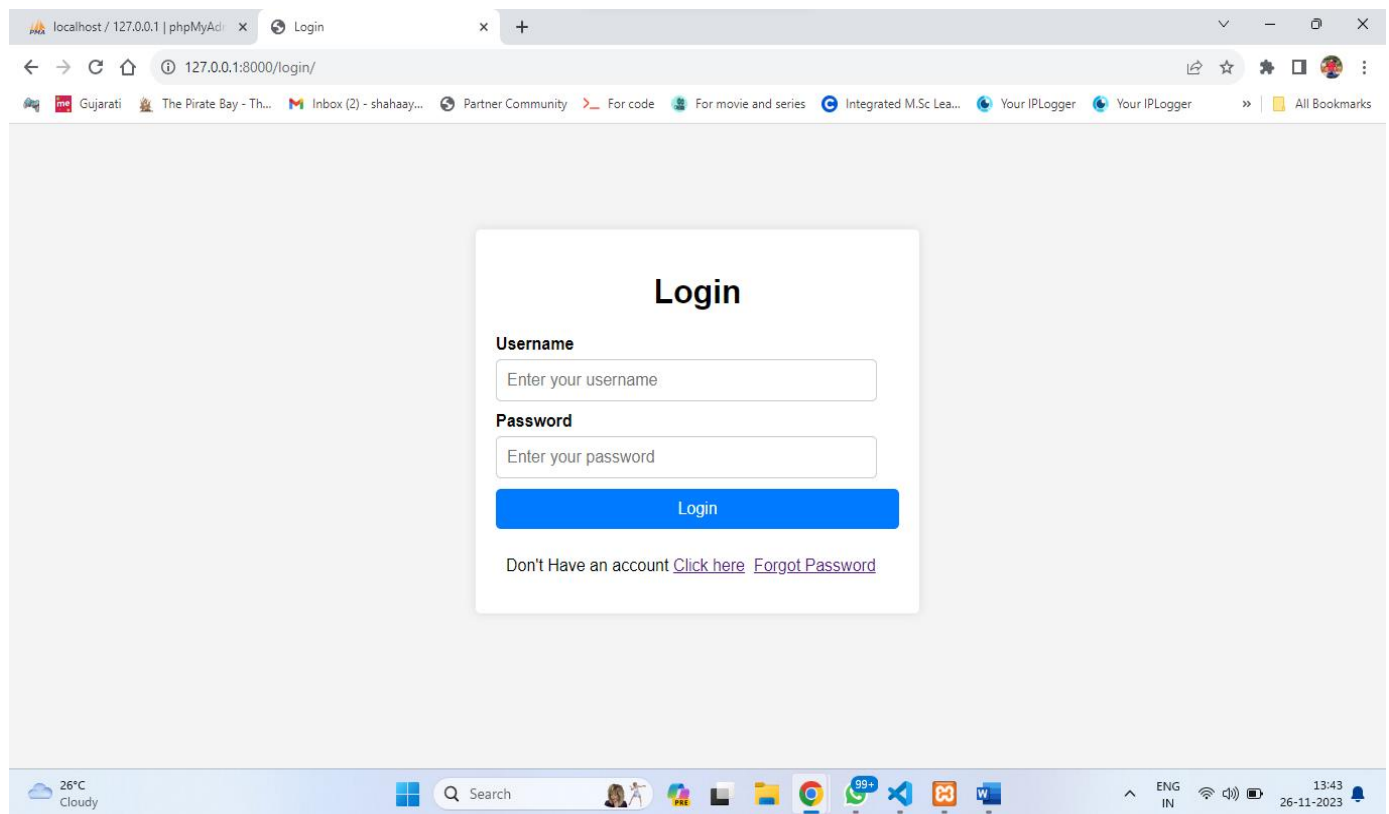
My team has been trying to go "agile" for few releases now, but being part of a large corporation hasn't exactly made it easy.

4.1 Coding Standards:

Dividing developers per module: Planning meetings and daily stand ups can become incredibly to everyone focus too much on their own modules. We tried that for one release, when management decided they will impose agile on the whole team and it will be completely their way. It as an absolute train wreck. If some of your team members prefer to work in isolation, let them (to a degree). If they prefer to work in pairs, let them do that. Make sure to let your people pick their own work.

4.2 Screenshots:





localhost / 127.0.0.1 | phpMyAdmin x Welcome to Project Manager Dashboard x +

127.0.0.1:8000/project_manager_dashboard/2/pm/pm@gmail.com/project_manager/python/

Gujarati The Pirate Bay - Th... Inbox (2) - shahaay... Partner Community For code For movie and series Integrated M.Sc Lea... Your IPLLogger Your IPLLogger All Bookmarks

Welcome to Project Manager Dashboard

Manager: pm

hi pm

- Logout

List of My Projects

- pro - Status: pending

Pending Update Status

Current Status:

List of Repository

Create Repository

Name	Description	Assigned To	Upload File	Upload Image	Action
a		user	repository_files/Screenshot_2023-11-24_194527.png	repository_images/Screenshot_2023-11-25_022839.png	

[View Tasks](#) [View Collaborator](#) [View Workspace](#)

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localhost / 127.0.0.1 | phpMyAdmin x Workspace List x Downloads x +

127.0.0.1:8000/workspaces/

Gujarati The Pirate Bay - Th... Inbox (2) - shahaay... Partner Community For code For movie and series Integrated M.Sc Lea... Your IPLLogger Your IPLLogger All Bookmarks

CollaboraShare Home Logout

Workspace List

Create Workspace Generate PDF Report Generate Text Report Generate Excel Report Generate CSV Report

Name	Description	Members	Categories	Active	Actions
------	-------------	---------	------------	--------	---------

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5. Agile Documentation

5.1 Project Charter

Agile Project Charter

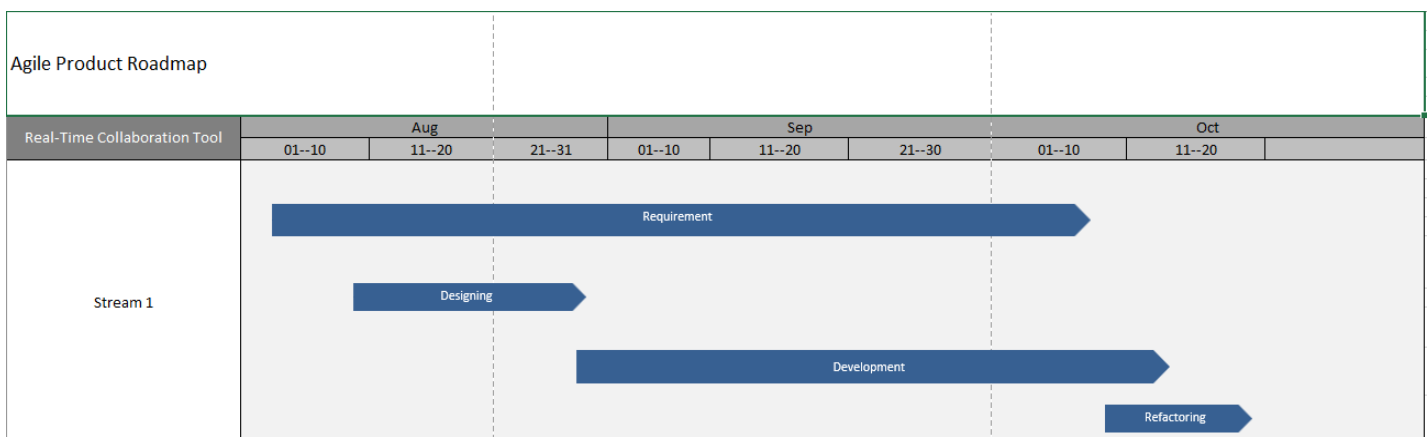
General Project Information

Project Name	Real-Time Collaboration Tool
Project Champion	Aayush Shah
Project Sponsor	LJ Institute Of Computer Application
Project Manager	Meet Shah
Stakeholders	Maharsh Poriya
Expected Start Date	1st Aug 2023
Expected Completion Date	20th Oct 2023

Project Details

Mission	Empower seamless, real-time collaboration for teams, enhancing productivity and creativity
Vision	To create a world where people can effortlessly connect and work together from anywhere, fostering a collaborative environment that transcends physical boundaries.
Scope	This tools have a bright future, with increasing demand for remote work and seamless teamwork
Date:	25th Oct 2023

5.2 Project Roadmap/Schedule



Real Time Collaboration Tool

5.3 Project Plan

Task Name	Start Date	End Date	Days	Staus	
Sprint 1				complete	
Registration & Login Form	01-08-2023	06-08-2023	6	complete	
Make Task CRUD Request for creating task	07-08-2023	11-08-2023	4	complete	
Provide search form based on task	12-08-2023	13-08-2023	1	complete	
Sprint 2				complete	
Designing Dashboard	14-08-2023	16-08-2023	2	complete	
Buld Upload in task CRUD	17-08-2023	18-08-2023	1	complete	
Make Collaborator CRUD Request for managing collaborator	19-08-2023	24-08-2023	5	complete	
Sprint 3				complete	
Make Workspace CRUD Request for managing workspace	25-08-2023	31-08-2023	6	complete	
Repost Generation of workspace	01-09-2023	03-09-2023	2	complete	
Forgot Password	04-09-2023	07-09-2023	3	complete	
Make Repository CRUD Request for managing Repository	08-09-2023	12-09-2023	4	complete	
Sprint 4				complete	
Update Profile	13-09-2023	15-09-2023	3	complete	
Changing of Design and Forms of the System	16-09-2023	20-09-2023	4	complete	
Make Feedback	21-09-2023	23-09-2023	2	complete	
View of Task	24-09-2023	26-09-2023	2	complete	
View of Collaborator	27-09-2023	29-09-2023	2	complete	
View of Workspace	30-09-2023	02-10-2023	2	complete	
visualize the usages of user	03-10-2023	10-10-2023	10	complete	
Refactor the code with new Technology	12-10-2023	19-10-2023	7	complete	
Responsible	Meet Shah				

5.4 Agile User Story

Agile User Story Template			
User Story ID	As a <type of user>	I want to <perform some task>	so that I can <achieve some goal>
1	Admin	View a status report from each team member	Ensure the project stays on track.
2	Project manager	Assign a task to end User	They can complete the task
3	User	Login	I can use the system as existing User
4	Admin	Manage Collaborator	I can control user activities and account activities
5	Project manager	Manage Tasks	I can control the user task and activities of task
6	User	Manage Profile	I can edit my profile account & review profile
7	Admin	View Feedback	I can view feedback of Users
8	Project manager	Manage Workspace	I can control the workspace of Users
9	User	Give Feedback	I can give feedback for system
10	User	View Tasks	I can view task that assign from the Project Manager

5.5 Agile Release Plan

Task Name	Start Date	End Date	Days	Staus	Release Date	Goal
Sprint 1				complete		
Registration & Login Form	01-08-2023	06-08-2023	6	complete	07-08-2023	user can Succesfully Login
Make Task CRUD Request for creating task	07-08-2023	11-08-2023	4	complete	12-08-2023	Manage task
Provide search form based on task	12-08-2023	13-08-2023	1	complete	14-08-2023	We can search the task
Sprint 2				complete		
Designing Dashboard	14-08-2023	16-08-2023	2	complete	18-08-2023	Looking beter view
Bulk Upload in task CRUD	17-08-2023	18-08-2023	1	complete	19-08-2023	We can use to uplaos bulk data via CSV
Make Collaborator CRUD Request for managing collaborator	19-08-2023	24-08-2023	5	complete	25-08-2023	Manage Collaborator
Sprint 3				complete		
Make Workspace CRUD Request for managing workspace	25-08-2023	31-08-2023	6	complete	31-08-2023	Manage Worksapce
Repost Generation of workspace	01-09-2023	03-09-2023	2	complete	03-09-2023	Generate report of detail of workspace
Forgot Password	04-09-2023	07-09-2023	3	complete	07-09-2023	We can forgot password for existing system
Make Repository CRUD Request for managing Repository	08-09-2023	12-09-2023	4	complete	12-09-2023	Manage Repository
Sprint 4				complete		
Update Profile	13-09-2023	15-09-2023	3	complete	15-09-2023	We can allow to edit profile of system
Changing of Design and Forms of the System	16-09-2023	20-09-2023	4	complete	20-09-2023	We can design for better look
Make Feedback	21-09-2023	23-09-2023	2	complete	23-09-2023	Make feedback
View of Task	24-09-2023	26-09-2023	2	complete	26-09-2023	View task details
View of Collaborator	27-09-2023	29-09-2023	2	complete	29-09-2023	View Collaborator details
View of Workspace	30-09-2023	02-10-2023	2	complete	02-10-2023	View Workspace details
Responsible	Aayush Shah					

5.6 Agile Sprint Backlog

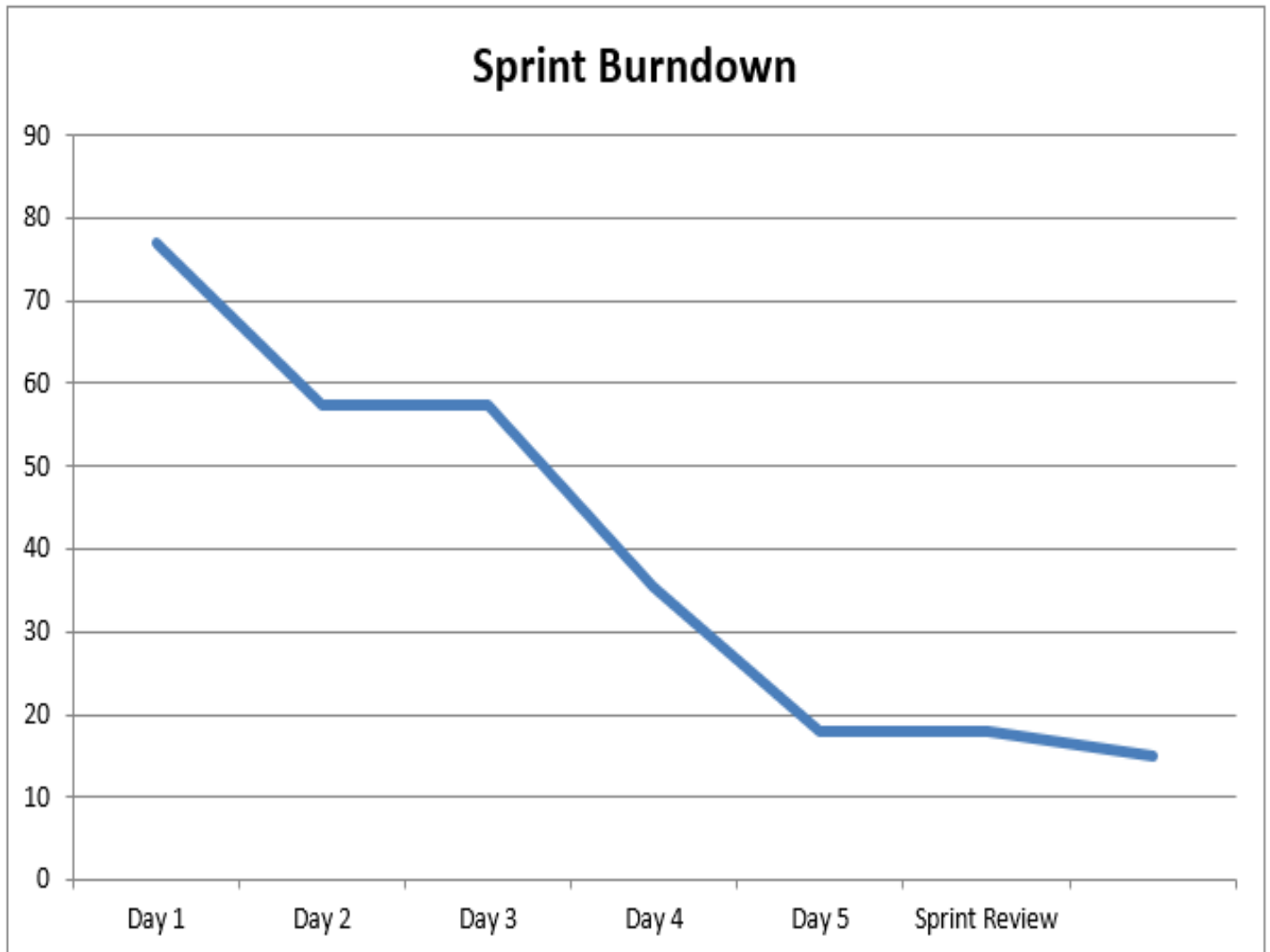
Sprint Backlog Template

Backlog Item	Story Points	Priority	Status	Original Estimate	Day 1	Day 2	Day 3	Day 4	Day 5	Sprint Review
Sprint 1	8									
Registration & Login Form	4	Medium	Done	7	5	3	0	0	0	0
Make Task CRUD Request for creating task	3	High	Done	3	1	1	5	0	1	0
Provide search form based on task	1	Medium	Done	1	4	5	3	0	0	1
Sprint 2	5									
Designing Dashboard	2	Low	Done	3	3	0.5	0.5	0	0	2
Buld Upload in task CRUD	1	Medium	Done	3	5	5	1	1	1	0
Make Collaborator CRUD Request for managing collaborator	2	High	Done	2	2	5	0	1	0	1
Sprint 3	8									
Make Workspace CRUD Request for managing workspace	4	High	Done	8	6	0	0	0	0	0
Report Generation of workspace	2	Medium	Pending	3	1	3	3	3	0	1
Forgot Password	1	Low	Done	6	1	5	3	1	1	0
Make Repository CRUD Request for managing Repository	1	High	Done	2	0.5	0	0	0	0	3
Sprint 4	16									
Update Profile	1	Medium	Done	9	4	2	2	1	1	0
Changing of Design and Forms of the System	3	Low	Done	6	6	3	3	3	1	1
Make Feedback	3	Low	Done	6	2	8	8	1	0	1
View of Task	2	Low	Done	0.5	0.5	0.5	0.5	0	0	0
View of Collaborator	1	Low	Done	2	1	1	1	0.5	1	1
View of Workspace	3	Low	Pending	6	6	6	0.5	3	9	0
Visualize the usages of user	1	High	Done	9	9	9	4	3	3	3
Refactor the code with new Technology	2	High	Done	0.5	0.5	0.5	1	0.5	0	1
Total	37		Done	77	57.5	57.5	35.5	18	18	15

5.7 Agile Test Plan

Test	Date	Action	Expected Result	Actual result	Pass?
1	02-08-2023	Logging	Get Collaboration Information	user redirect to different page	pass
2	04-08-2023	Login successfully	Display message according to login successful	successful login	pass
3	09-08-2023	Task	View task of specific user	View task of specific user	pass
4	09-09-2023	Repository	View repository of specific user	View repository of specific user	pass
5	21-08-2023	Collaborator	View collaborator of specific user	View collaborator of specific user	pass
6	26-08-2023	Workspace	View workspace of specific user	View workspace of specific user	pass
7	02-09-2023	Generate Report	Generate report of workspace	Error in XML generate	fail
8	22-09-2023	Give Feeback	Give Feeback for system by user	Give Feeback for system by user	pass
9	14-09-2023	Update Profile	Edit profile for user	Edit profile for user	pass
10	06-09-2023	Forgot Password	Reset the password	otp not came in mail	fail

5.8 Earned-value and burn charts



6. PROPOSED ENHANCEMENTS

- Does not provide editor in workspace

7. CONCLUSION

In conclusion, a real-time collaboration tool is a valuable solution that allows people to work together on projects, documents, and tasks in real-time. It enhances productivity, fosters teamwork, and simplifies communication by enabling users to collaborate seamlessly, regardless of their physical locations. With features like simultaneous editing, chat, and file sharing, it streamlines work processes and promotes efficiency. Real-time collaboration tools have become essential for modern work environments, making it easier for teams to achieve their goals and deliver results faster. Whether for business, education, or personal use, these tools are a game-changer in today's interconnected world.

8. BIBLIOGRAPHY

- www.github.com